PSID Crisis Module: What the Optimized Design Delivers

Validated portfolio metrics from the final ranked questionnaire workflow

The final module keeps the questionnaire inside the 30-minute field constraint while preserving the highest-value, most portable crisis questions.

52 -> 28

Candidate questions narrowed to deployable module

29.17 / 30 min

Selected module duration under hard interview cap

2.452 / 7.809

Average / maximum Pi enhanced-priority score

14 recommendations

Documentation-backed outputs: 6 generic core + 8 crisis-specific

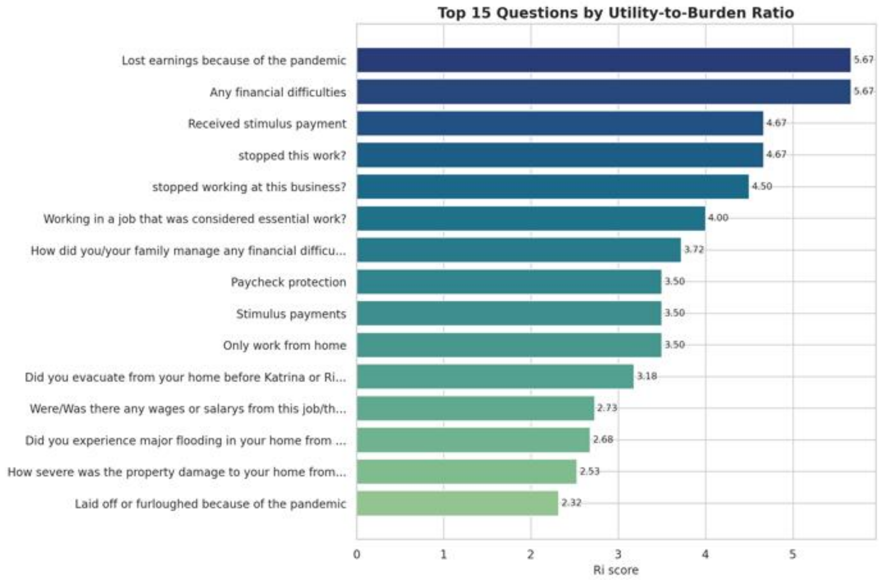
- Selection outcome: 28 questions retained from 52 ranked candidates, consuming 97.2% of the 30-minute ceiling without exceeding it.

Sources: psid_artifact_summary.json and generate_psid_artifacts.py

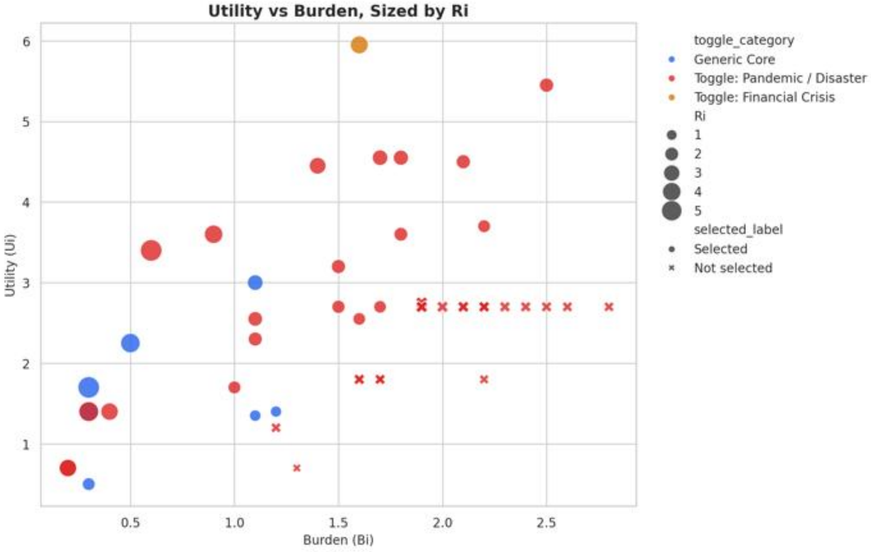
Highest-Value Questions and the Utility-Burden Tradeoff

The validated analysis figures are embedded directly below as slide images

Top-ranked questions by utility-to-burden ratio



Utility versus burden profile across the crisis corpus



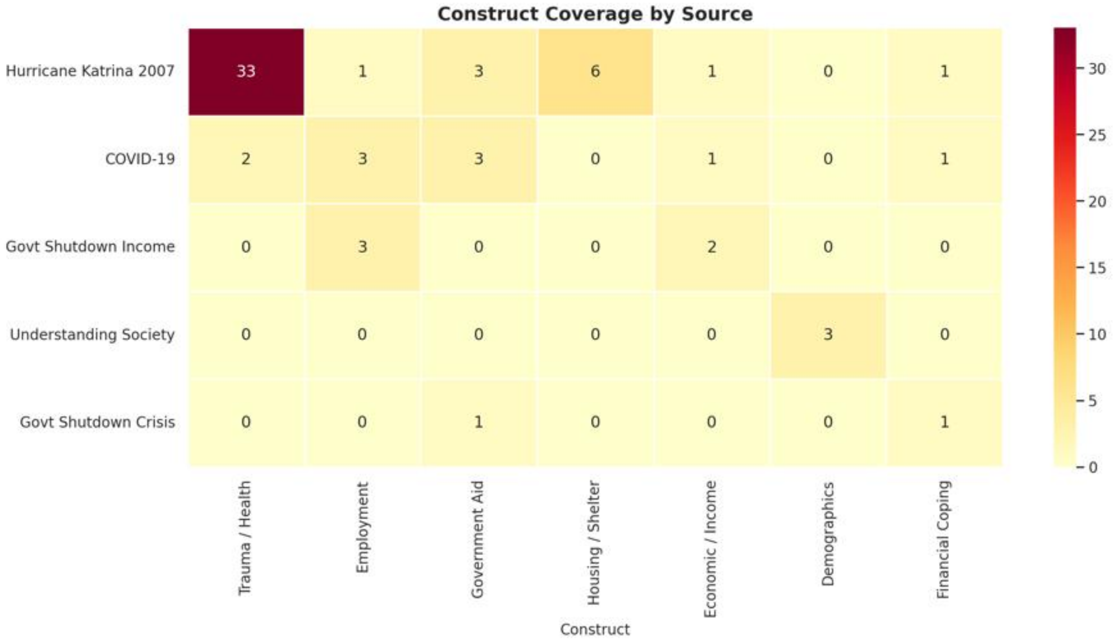
- Better questions appear higher in value and lower in burden; the scatter makes that tradeoff intuitive for nontechnical audiences.

Embedded figures: fig_top_ranked_questions.png and fig_utility_vs_burden.png

Coverage Across Constructs, Sources, and Time Budget

Breadth of coverage, evidence sources, and the final time allocation are shown together

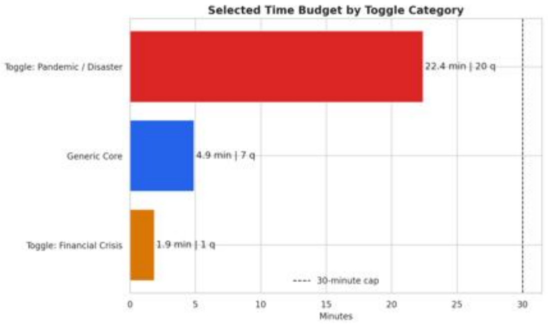
Construct coverage by source



Selected source contribution

- Hurricane Katrina 2007: 13 selected
- COVID-19: 8 selected
- Govt Shutdown Income: 3 selected
- Understanding Society: 3 selected
- Govt Shutdown Crisis: 1 selected

Time budget by toggle category



Embedded figures: fig_construct_heatmap.png and fig_time_budget.png

Metrics That Are Easy to Explain to Stakeholders

The selection logic is rigorous, but the outputs are simple to communicate

Core performance metrics

- 52 total ranked questions; 28 selected for the deployable module.
- 29.17 selected minutes versus 68.95 minutes for the full corpus.
- Average Ri: 2.079; maximum Ri: 5.667.
- Average Pi: 2.452; maximum Pi: 7.809.

Simple interpretation

- Ri answers a straightforward question: how much thematic value do we get per unit of burden?
- Pi improves that score by rewarding rarer wording, richer construct coverage, and deployable phrasing while penalizing redundancy.
- The selected module ends near the 30-minute cap because the workflow fills the available interview time with the highest-priority items first.

Recommendation outputs

6

Generic-core recommendations

8

Crisis-specific recommendations

Each recommendation is linked to documentation-backed patterns and supported by the highest-scoring matching questions in the corpus.

Metrics derived from psid_artifact_summary.json and recommendation generation in generate_psid_artifacts.py

Computational Model Architecture

From raw PSID source material to deployable, evidence-backed questionnaire artifacts

1. Data Input

- 5 PSID-related sources
- 52 integrated candidate questions
- Authoritative base CSV for ranking

2. NLP Processing

- Keyword parsing and tagging
- Construct extraction
- Question-level feature assembly

3. Augmented Scoring

- TF-IDF rarity
- Cosine redundancy penalty
- Construct richness and portability bonus

4. Selection Engine

- Rank by Pi descending
- Greedy fit to 30-minute cap
- Deployable wording and recommendations

5. Artifact Output

- Final CSV with 25 columns
- Dashboard data payload
- PNG figures, report HTML, notebook outputs

Enhanced priority formula

$$P_i = U^* / [B_i \times (1 + 0.65 \times \text{redundancy})]$$

$$U^* = U \times (1 + 0.32 \times \text{idf} + 0.24 \times \text{construct} + 0.12 \times \text{richness} + \text{portability})$$

$$B_i = \max(0.10 \times \text{word\_count} + 0.20 \times \text{complexity}, 0.1)$$

Construct priority weights

- Trauma/Health 0.58
- Housing/Shelter 0.55
- Government Aid 0.48
- Financial Coping 0.45
- Employment 0.42
- Economic/Income 0.40
- Demographics 0.14

Implemented in generate_psid_artifacts.py via build_all() with TF-IDF, redundancy control, and recommendation generation